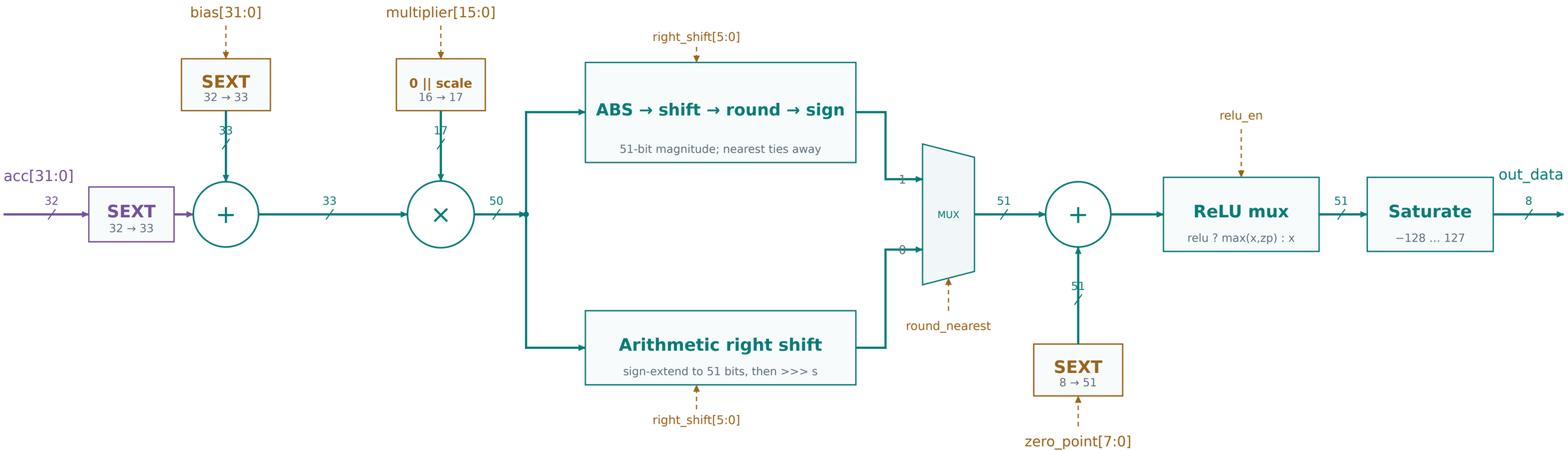
Affine requantizer — widened integer datapath

Default widths shown: ACC_W=32, OUT_W=8. All blocks are combinational; per-column configuration is held separately.



Bias addition is 33 bits, scale multiplication is 50 bits, and shift/round/zero-point operations use 51-bit guarded values.
The final clamp alone reduces precision to OUT_W. ReLU clamps at output zero_point; input zero-point subtraction is outside this datapath.